TP1: Linear models for binary classification

1- Implement the following algorithms :
   - Perceptron learning algorithm.
   - Pocket learning algorithm.
   - Delta rule learning algorithm.

2- Use or create one data set of your choice, having nonlinearly separable data due to noise, then visualize it. Note that $d = 2$ and $y = \{-1, +1\}$.

3- Apply the aforementioned algorithms to your data (for Pocket and delta use $T_{max} = 100$).

4- For each of the algorithms provide the graph of the empirical error evolution.

5- Compare the three algorithms in terms of:
   - Number of iterations.
   - Accuracy.
   - Empirical error evolution.

(Comment your results)